

Research paper

Leveraging generative adversarial networks for data augmentation to improve fault detection in wind turbines with imbalanced data

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ABSTRACT

In the realm of modern power systems, wind turbines (WTs) have gained significant environmental advantages, making them pivotal in the expanding landscape of wind power generation. The paramount task of ensuring the effective monitoring and fault classification of WTs is indispensable for the stability of wind farm systems. Nevertheless, a formidable challenge arises from the need for more fault information gleaned from small fault data samples, resulting in data insufficiency and imbalance issues. The consequential impact of such imbalances on fault detection accuracy underscores the critical nature of the concern in wind turbine fault diagnostics. Notably, the disparity in dataset sizes among faults and the uneven distribution of fault classes pose a substantial obstacle in the realm of fault detection for wind turbines. This challenge is compounded by the time-intensive process required to amass sufficient fault data, setting it apart from more typical scenarios in the domain of wind turbines. This paper introduces an innovative approach employing Generative Adversarial Networks (GAN) for synthetic fault data generation to address these challenges effectively. This paper utilizes a Wasserstein Conditional Generative Adversarial Network (WC-GAN), which replaces the KL divergence in CGAN with the Wasserstein distance to rectify data imbalances by generating synthetic fault samples for wind turbine fault classification. This strategic manoeuvre balances the class distribution and enhances fault classification accuracy. Incorporating conditional data generation contributes to training stability and sample quality while utilising Wasserstein distance ensures a faster convergence rate. Experimental validation conducted on Supervisory Control and Data Acquisition (SCADA) data for fault classification of wind turbines underscores the superiority of our method over other approaches, primarily attributed to the quality of conditionally generated samples. In essence, the proposed approach adeptly tackles the challenge of imbalanced samples by generating high-quality synthetic fault data, thereby elevating the efficacy of wind turbine fault classification.

1. Introduction

The environmental problem is raised by the ongoing use of fossil fuels and rising pollution. The issue of wind turbine problems has received much attention recently as large-scale wind turbines have been put into service. Due to the benefits of being quiet, pollution-free, infinitely renewable, and clean, several nations have heavily concentrated on and implemented renewable energy, such as wind turbines [1]. According to the report by enegytrackerasia, by 2030, South Korea wants to produce 20% of its energy using renewable sources [2]. More than twelve wind power plants, including Jeju Island, are operating in South

Korea to generate renewable energy. A massive disaster or significant time and financial losses could result from an unforeseen equipment failure during operation [3]. Early fault identification and diagnosis ensure electric devices function normally and generate more significant economic benefits. As a result, a fault detection system must be created to produce precise and immediate diagnostics. Additionally, the operator needs to use the proposed intelligent technique to ensure accurate fault identification, develop a predictive maintenance strategy, and have correct operation planning. Early fault identification and diagnosis ensure electric devices function normally and generate more significant economic benefits. As a result, a fault detection system must

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be created to produce precise and immediate diagnostics. Additionally, the operator must use the proposed intelligent technique to ensure accurate fault identification, develop a predictive maintenance strategy, and have correct operation planning [4,5]. Remotely monitoring component health is even more critical in the WTs than in other industries. Wind turbines are typically designed to function independently in remote or offshore locations, where continuous visual inspections may be impractical. Unexpected failures in WTs can be more expensive; necessary corrective repair may consume a sizable percentage of the turbine's yearly maintenance budget. Wind turbines appear to have highly irregular loads compared to other rotating machines because of fluctuating and turbulent wind conditions. As a result, components might experience considerable stress throughout their lifetime [6]. By adding extra sensors, Condition Monitoring Systems (CMSs) on wind turbines offer detailed prognoses on turbine sub-assemblies. The turbine already has several sensors in the SCADA system. In recent years, SCADA data has developed quickly. The SCADA data-based [7] techniques are economical because no extra sensors are required. A significant amount of data may be gathered thanks to the development of information technology. Artificial intelligence (AI)-based approaches [8,9] are better for use because of the data. Artificial intelligence-based fault detection and WT modeling techniques have advanced significantly. Machine learning (ML)-based fault classification frequently requires much historical data to train the model [10–12]. The issue with fault detection based on machine learning is that the model must reliably identify when a new defect develops. One of the biggest problems with fault classification in wind turbines is data imbalance [13–16]. Numerous studies have been conducted on the significance of imbalance problems in machine learning, and numerous researchers have been working to find solutions by using various data resampling approaches and synthetic data generation with GAN [17–20]. A class imbalance in the dataset, for instance, might significantly harm classifier performance by showing a bias in predictions in favour of the majority class.

Imbalanced data classification is a well-known challenge in machine learning, especially in wind turbine fault classification. Current methods, including resampling and algorithmic-level approaches, often cannot effectively handle the nuances of small class sampling. In this study, we propose an improved WCGAN to generate synthetic error samples, balance the dataset, and improve classification accuracy.

2. Literature review

2.1. Data imbalance in the application domain

There is a great deal of complexity involved in classifying uneven data. A resampling technique, which alters distributions at the data level, is aimed at reaching statistical equilibrium. It is important to keep in mind that oversampling can result in overfitting, while undersampling can result in discarding valuable majority-class data. Data imbalance in wind turbine SCADA systems poses unique challenges. Unlike other fields, errors in SCADA data sets are very rare and only a part of the overall data, which significantly biases the classifier compared to normal working samples. Traditional techniques such as oversampling and undersampling often exacerbate these problems by overfitting a small number of classes or removing valuable information in a large number of classes. Recent advances such as GAN and SMOTE attempt to address these issues. However, it is often not possible to generate the sample diversity needed for complex multi-class data sets such as SCADA. The authors [21] built a DNN-based methodology that will generate synthetic data samples to control imbalanced classes using GAN and Synthetic Minority Oversampling Technique (SMOTE) [22] to oversample the data. The GAN creates more synthetic samples to balance and expand the training dataset and improve overall classification accuracy. To address the issue of imbalanced data, synthetic over-sampling methods often try to create fresh data samples of minority classes. Data augmentation is a method for resolving the problem of data imbalance,

which refers to a dataset having an excessive number of classes, especially in case studies involving wind turbines. Since GAN has a high degree of accuracy when it comes to fitting data, it is commonly utilized for creating synthetic data and overcoming unbalanced learning difficulties. Various domains that address issues related to data scarcity and imbalance can benefit from the application of synthetic data generation using GANs. As demonstrated by the work of Chatterjee et al. [23–25], they demonstrate improved model robustness and reliability using GAN-based technologies in their work on forecasting demand for electric vehicles and renewable energy systems. The study by Hazra et al. [26] presents ensemble methods combined with synthetic data generation as a method of predicting fermentation processes in traditional beverages. It can be challenging to deal with data imbalance issues in machine learning tasks, especially when addressing domains where the data distribution is naturally skewed. Recently, work developed by Chatterjee et al. [27] tackled these challenges using advanced resampling strategies and hybrid ensemble methods, which significantly improved classification performance in imbalanced datasets for wind turbines and renewable energy applications.

Dixit and Verma [28] used a conditional variational autoencoder (CVAE) to create synthetic data samples to address the issue of small sample sizes in the training dataset to address the data imbalance problem in monitoring rotary machines. Using artificial data created by their suggested approach, the classification accuracy increased by 63%. To produce accurate one-dimensional synthetic raw samples from mechanical sensor inputs and learn from their signals, Shao et al. [29] built an auxiliary classifier GAN (ACGAN)-based framework. Cabrera et al. [30] proposed an improved generative model-based GAN method to overcome extreme imbalance data and random forest classifier for classification tasks. Lui et al. [31] suggested a sparse semi-supervised GAN for defect diagnosis employing a large amount of unlabeled data to increase accuracy. They explained how cost-sensitive learning and boundary-seeking loss are used to stabilize training and produce more useful data. An unsupervised fault diagnostic technique using a Quick Self-attention Convolutional Generative Adversarial Network (QSCGAN) was proposed by Wan et al. [32]. To improve the training dataset and the classifier's classification accuracy, the pair networks generator and discriminator first learn distribution to the real data before generating raw mechanical signals. A deep autoencoder (DA) model was presented in another work by Wang et al. [33] to use SCADA data to predict impending blade breaks. The authors [34], proposed a conditional GAN (CGAN) and a two-dimensional convolutional neural network for identifying bearing defects using a small sample size. CGAN is used to train the generator and discriminator in confrontational learning to generate new samples with the same data distribution while considering the problem of insufficient sample data. The research authors [35] suggested a unique conditional auxiliary classifier GAN framework in combination with model agnostic meta-learning (MAML) to learn and create the task of synthetic sample production utilizing the limited training dataset. GAN is beneficial for producing high-quality samples. Recent research has explored IoT and machine learning-based approaches to improve energy management and anomaly detection. In the work by Qayyum et al. [36] a peer-to-peer nanogrid network with the aim of achieving two distinct objectives through the design of an optimal energy trading service architecture. In another work [37] IoT-enabled nanogrid energy trading architecture that optimizes energy trading costs and ESS power utilization using particle swarm optimization (PSO) and enhances performance through virtual operation with IoT task orchestration technology. The work by Khan et al. [38] in order to improve accuracy and achieve high detection rates, an optimized ensemble intrusion detection system (OE-IDS) is implemented using AutoML and a soft voting method. In another work [39] to detect and eliminate clinical data anomalies, we applied an enhanced hybrid abnormal data detection strategy that combined heuristic k-means clustering with a stochastic interquartile range method.

2.2. Challenges in state-of-the-art augmentation techniques

State-of-the-art augmentation techniques have significant limitations when it comes to effectively addressing data imbalance. Data imbalance can be effectively addressed by state-of-the-art augmentation techniques, but they are not without critical limitations. Even with their promise, state-of-the-art augmentation techniques are not able to effectively address data imbalance. It is common for GANs to suffer from mode collapse, thereby reducing diversity of synthetic samples [40]. CVAE (Conditional VAE) can generate specific samples, but often requires extensive computing resources and fine tuning to maintain stability. For wind turbine data sets, these methods have difficulty generating realistic fault data due to the multidimensional and temporal nature of the SCADA data. To address these challenges, the proposed WCGAN incorporates Wasserstein distance and gradient penalty to stabilize training and generate fault samples that are of high quality. A WT is a multi-component, multi-subsystem complex system. Other techniques of SMOTE oversampling variants were also put forth to enhance the data-generating effect, such as borderlineSMOTE [41], which only oversamples the minority examples close to the borderline, and MSMOTE [42], which is modified to eliminate noise samples through adaptive mediation. He et al. [43] proposed the adaptive synthetic sampling approach Adaptive Synthetic(ADASYN). While it generates distinct samples based on an estimate of the local distribution of the class to be oversampled, it is comparable to SMOTE.

2.3. Research gaps and contributions

The issue of imbalanced SCADA data from wind turbines has not been adequately addressed by current research, and GAN-based approaches have been found to lack validation against highly biased real-world data sets. In this study, we address this gap by implementing a WCGAN tailored to SCADA data, generating high-quality synthetic defect samples, and integrating them with real data to improve the accuracy of defect detection. The contributions of this study are the following. We developed a WCGAN model to solve the unique problem of SCADA data imbalance. Integrate real and synthetic error samples to improve classifier performance. Compared to traditional dilation and grading techniques, it provides better results.

The primary contributions of this study are as follows:

- This study tries to identify faults in the wind turbines' publically available data. This paper proposes a synthetic fault data generation technique for WT fault classification. This technique can solve imbalanced data issues, allowing ML-based techniques to be effectively trained with a smaller number of real fault samples.
- We utilize Conditional GANs (CGAN) to develop both the generator and discriminator models, tailored to the characteristics of SCADA data. The improved CGAN is more suitable for generating fault samples, allowing for the generation of preferred fault samples by assigning different class labels.
- We use a technique that mixes actual fault samples and synthetic fault samples. The generated fault data is combined with real-world data and used to train the fault-detection model so that its performance can be enhanced.
- In this paper, we evaluate our proposed framework on a real dataset to show its effectiveness. The results of the experimental inspection show that our model can detect faults successfully and generate high-quality synthetic data, which can both solve the issue of imbalance and improve the classification effect.

The paper unfolds as follows: Section 2 delves into a comprehensive explanation of the proposed method. Following that, Section 3 highlights the efficacy of the suggested technique. The paper concludes in Section 4.

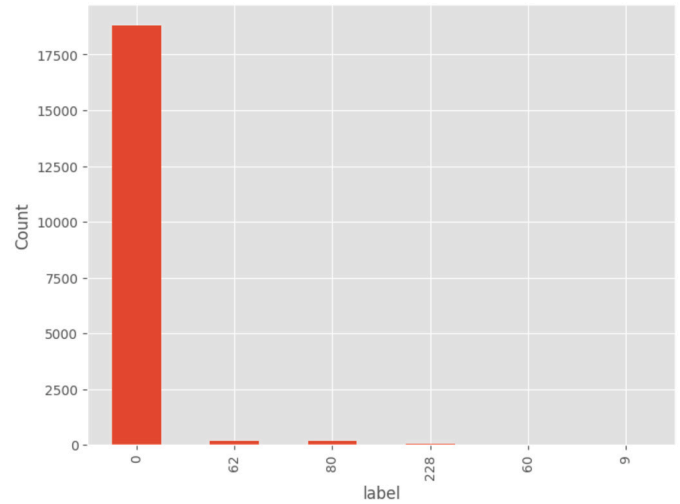


Fig. 1. Visualization of number of instances for each class including no-fault and five faults.

3. Case study and experimental setup

In this section, we describe in detailed the case study and the experimental setup used to validate our approach. It is discussed in depth how the data acquisition and pre-processing steps were performed for this study. Furthermore, we explain the post-processing process used for the generation of synthetic fault samples as well as the evaluation methodology for assessing fault classification model effectiveness.

3.1. Data acquisition and preprocessing

The data used in this study was obtained from a direct drive wind turbine powering a sizeable biomedical device manufacturing facility near a beach in southern Ireland [44]. Before proceeding to the data preprocessing stage, we combined SCADA and status data for the experiment, creating the final dataset. When wind supply is normal, and weather conditions are good, the chances of a fault occurring would be less. For this reason, the number of faults in wind turbines is much less than in no-fault or when the wind turbine is operational. 10-minute operational data is provided by the wind turbine SCADA system. Table 1 contains a sample of the 19208 rows and 62 columns of complete SCADA data. There is an 11-month period of mixed data, from May 2014 to April 2015, which contains no-fault and all fault labels in the dataset. Fig. 1 shows the number of instances for each class, including no-fault and five faults. For this experiment, we used the fault and no-fault data that SCADA data has retrieved. The label column contains six labels, with no-fault data represented by label 0 and fault data defined by 80, 228, 60, 62, and 9. We can determine that no-fault (label 0) is much higher than the other five fault labels. However, there is a big gap between the fault samples, which leads to imbalanced class issues.

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The number of instances of different faults is summarized in Table 2. Where excitation errors (80) are issues with the generator's excitation system; feeding faults (62) are problems with the turbine's power feeder cables; malfunctioning air cooling (228) is a problem with the turbine's air circulation and internal temperature circulation; fault (60) is a mains failure and start delay. Generator overheating is a generator heating

Table 1
A small sample of experimental original public data.

Time	WEC: ava. windspeed	WEC : max. windspeed	WEC : min. windspeed	WEC: ava. Rotation		CS101 : Tower temp.	CS101 : Control cabinet temp.	CS101 : Transformer temp.	RTU: ava. Setpoint 1	label
2014-05-01 08:50:00	4.1	9.5	0.8	0.00		14	23	33	2501	0
2014-05-01 11:30:00	9.3	12.6	2.6	0.00		14	17	30	2501	0
2014-05-01 11:30:00	9.3	12.6	2.6	0.00		14	17	30	2501	0
.....										
2014-10-26 19:00:00	11.6	17.0	8.1	9.98		25	33	17	3050	9
2015-01-07 19:10:00	12.2	20.1	2.6	8.53		13	21	31	3050	9
2015-01-07 19:20:00	8.1	16.1	2.5	2.74		13	21	31	3050	9

Table 2
Combination of frequently occurring faults, listed by status code, and number of instances.

Fault	Status Code	No. of instances
Feeding Fault	62	182
Excitation Error	80	168
Malfunction Air Cooling	228	40
Generator Heating Fault	9	12
Mains failure and start delay	60	20

fault (number 9). We have chosen five types of wind turbine faults for our investigation data. Fig. 2 shows the count of all five fault status readings for each selected class, “fault 62, fault 80, fault 228, fault 9, and fault 60”, which is the basis for classification, and the pie chart shows the class instances percentage wish. In particular, power faults are represented by feeding fault 62, which has 182 instances. The generator excitation system issues are represented by excitation error 80, which has 168 instances. Generator overheating is represented by generator heating defect number 9, which has 12 instances. Malfunction air cooling 228 refers to issues with the airflow and internal temperature circulation of the turbine, which has 40 instances. The mains failure number 60 denotes problems with the turbine’s mains electrical supply, which has 20 instances. There needs to be a better balance between the fault classes. The stator is one of the most important electrical parts of the wind turbine. In Fig. 3, we present the visualization of the Stator temperature for both the original SCADA data and the data encompassing all faults. The red line depicts the Stator temperature, while the blue line represents the occurrences of all faults. In Fig. 4, the red line highlights a specific area where faults were observed. Remarkably, the majority of defects were recorded prior to 2015. To focus on this critical period, we truncated the dataset in mid-January 2015 and omitted the data from February to April 2015. The resultant plot, represented by the blue line, illustrates the main experiment’s fault samples.

3.1.1. Data preprocessing-Step1

In the first step from the original data, the “Time” column, which holds the date and time value together, we extracted the hour, minute, month, year, and day of the month from the time column. So that the time column can be dropped to be further compatible with the GAN model.

3.1.2. Data preprocessing-Step2

In this experiment, we selected only fault labels that are significantly less on the original data. The data begins with all no-fault (label 0), from 18789 to 19208 rows in the original data containing 62, 80, 9, 228, and 60, only those fault labels. A total of 420 rows and 62 columns are selected, including the fault labels, except the no-fault label, which will be used further for training the GAN model. The sample instances used in this experimental study of different fault labels are summarized in Table 3. Fig. 5 indicates the number of instances present for each five fault classes.

Fig. 6 portrays the distribution of data pertaining to five distinct faults. The x-axis represents wind speed in meters per second (m/s), while the y-axis signifies actual power in kilowatts (KW). Each data point is represented by a colored dot, with red indicating a fault number of 62, yellow denoting a fault number of 9, purple representing a fault number of 228, blue signifying a fault number of 80, and ash gray highlighting a fault number of 60. This visual presentation enables a comprehensive examination of the fault distribution concerning wind speed and actual power values.

4. Methods

Established models for anomaly detection encompass various methodologies, each with its strengths and weaknesses. Statistical methods offer simplicity and efficiency but struggle with complex anomalies. Machine learning approaches provide flexibility and adaptation, but require substantial labeled data for training. Deep learning models excel at learning complex patterns but demand large datasets and substantial computational resources. Ensemble methods combine models for improved accuracy but can be computationally expensive. Unsupervised learning approaches eliminate the need for labeled data but may lead to false positives. Time-series analysis techniques are effective for temporal data, but may lack generalizability. In contrast, our proposed model addresses the challenge of imbalanced data in wind turbine fault classification. It leverages a CGAN to generate synthetic fault data. This method enables effective training with a smaller number of real fault samples, thereby enhancing fault detection accuracy and balancing the data imbalance. Additionally, it maintains computational efficiency. By integrating both real and synthetic fault samples for model training and testing with unseen data, our approach demonstrates improved fault classification performance. This effectively leverages the advantages of both real and generated data to enhance anomaly detection in wind turbine systems.

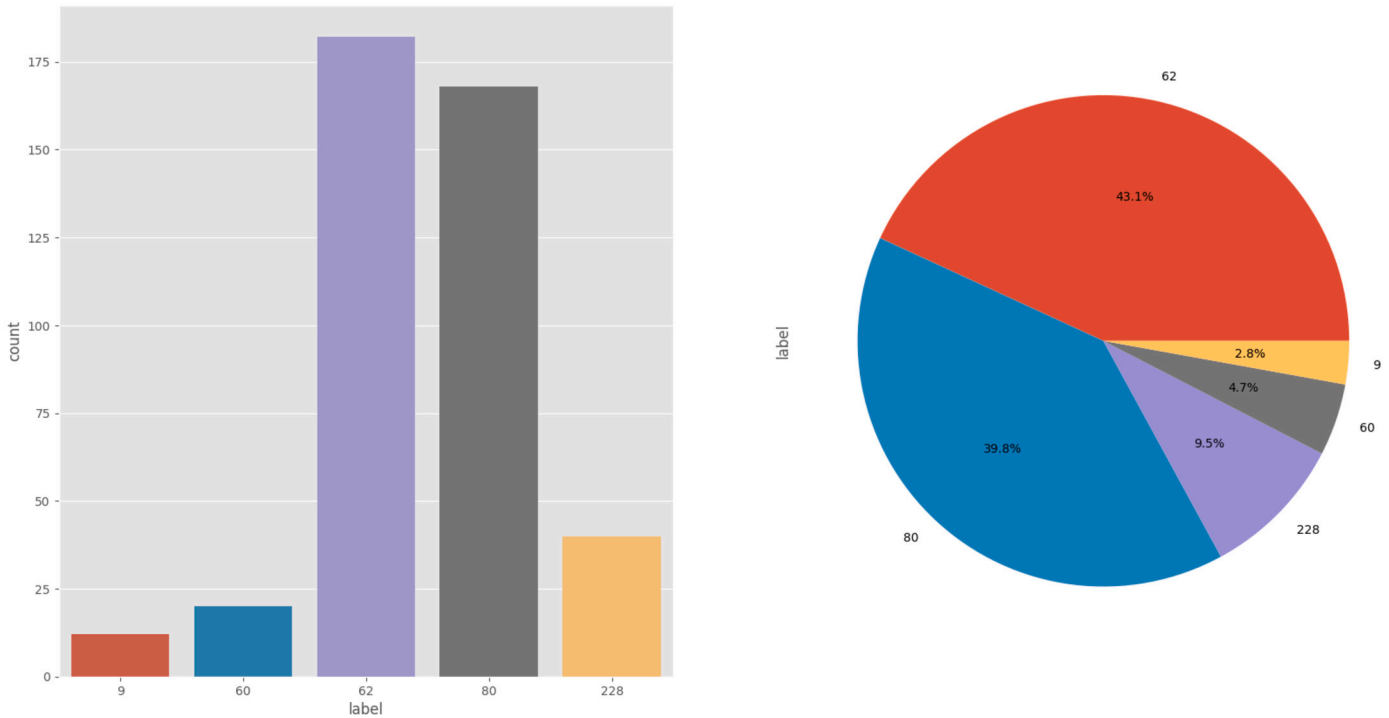


Fig. 2. Visualization of instances across the five fault classes and a pie diagram to show the percentage-wise data distribution.

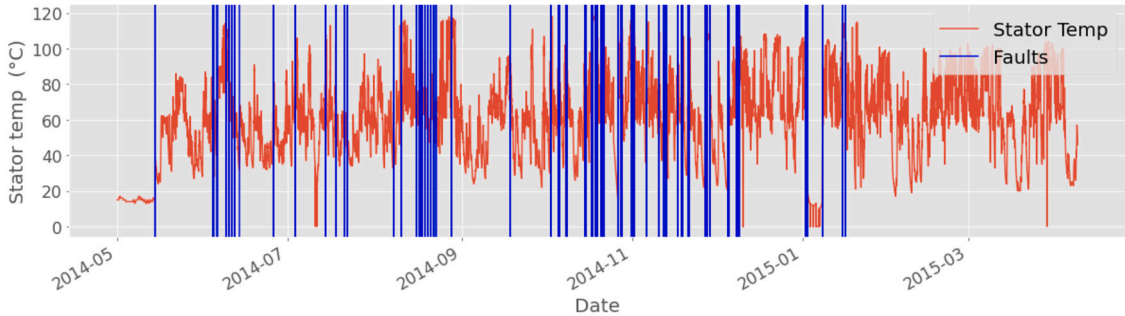


Fig. 3. The visualization highlights the faults along with their corresponding stator temperature values.

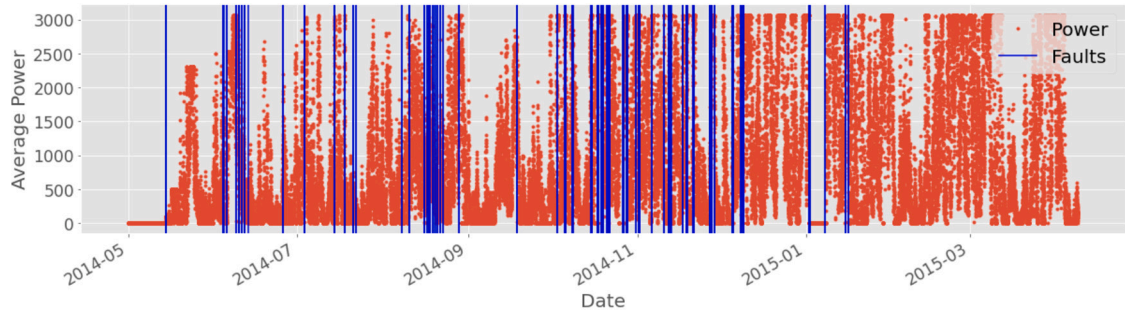


Fig. 4. The visualization highlights the faults along with their corresponding power values.

Table 3
Visualization of the fault data that contains only fault samples.

Time	WEC: ava. windspeed	WEC : max. windspeed	WEC : min. windspeed	WEC: ava. Rotation		CS101 : Tower temp.	CS101 : Control cabinet temp.	CS101 : Transformer temp.	RTU: ava. Setpoint 1	label
2014-07-14 12:30:00	12.8	17.4	5.8	14.04		33	40	57	3050	62
2014-07-14 12:40:00	12.9	20.2	5.8	9.27		32	41	59	3050	62
2014-07-18 01:10:00	11.2	12.8	9.4	14.05		33	40	54	3050	62
.....										
2014-10-26 19:10:00	11.6	17.0	8.1	9.98		25	33	17	3050	9
2015-01-07 19:10:00	12.2	20.1	2.6	8.53		13	21	31	3050	9
2015-01-07 19:20:00	8.1	16.1	2.5	2.74		13	21	31	3050	9

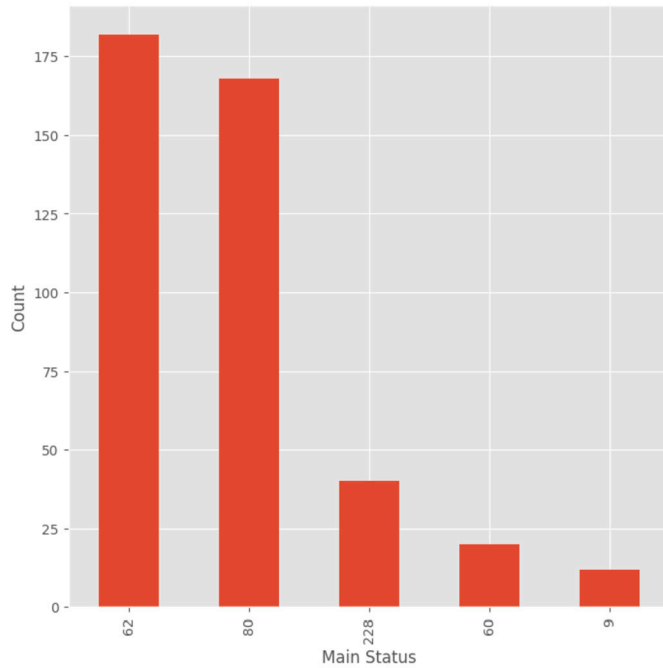


Fig. 5. Visualization of number of instances for each five fault classes.

The methods used for fault classification of wind turbines using an imbalanced dataset are described in this section. The methodology's primary goal is to provide a sizable volume of fault data to give adequate samples for fault categorization. The procedure is broken down into the following phases, as shown in Fig. 7. The experimental SCADA and status data are cleaned and smoothed before a detailed description of the data analysis. The study advises employing the data-level strategy to get a more evenly distributed class distribution in the dataset. The recommended method takes advantage of data augmentation based on GAN. Data were then prepared before being fed into the GAN model to create synthetic fault samples. The generation of synthetic fault data is then explained using the proposed WCGAN model to address a suitable learning environment to settle data imbalance issues. We undertake some data post-processing after data generation, which is covered in the next section. Finally, ensemble-based classification models were investigated to

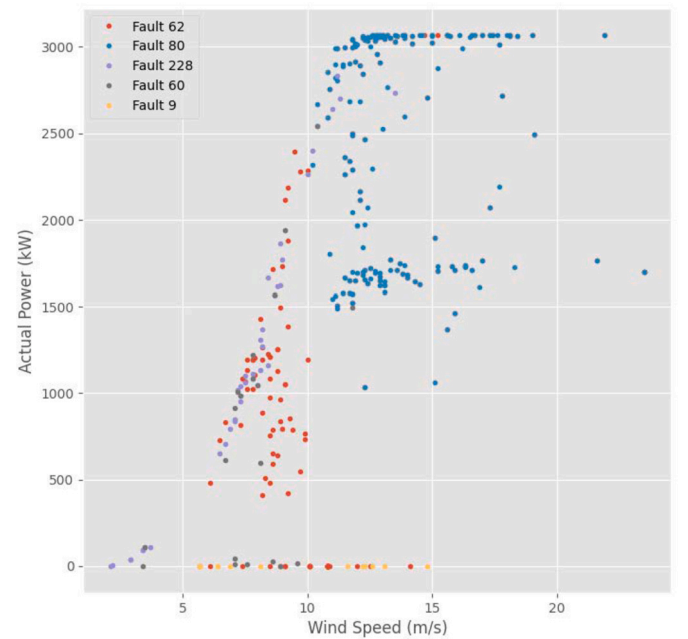


Fig. 6. Distribution of data for five distinct faults based on wind speed (m/s) and actual power (KW).

create a fault detection model in WT. We outline an ensemble strategy for the fault classification models used in this investigation.

4.1. Generative adversarial networks

The method proposed by [45], where two networks engaged in a game of competition make up most of the GAN. A GAN technique can generate more accurate and realistic samples compared to other methods. Two distinct sub-networks for a GAN compete with one another and improve through backpropagation. Training the GAN model to use an adversarial process and learn more evenly distributed data can help solve the problem of data imbalance. The GAN structure comprises a generating network (G) and a discriminator network (D). G's purpose is to produce fake data that seems like the real thing. D's function determines whether the input data are actual or generated counterfeit samples. The discriminator should provide 1, where $D(x) = 1$ if the in-

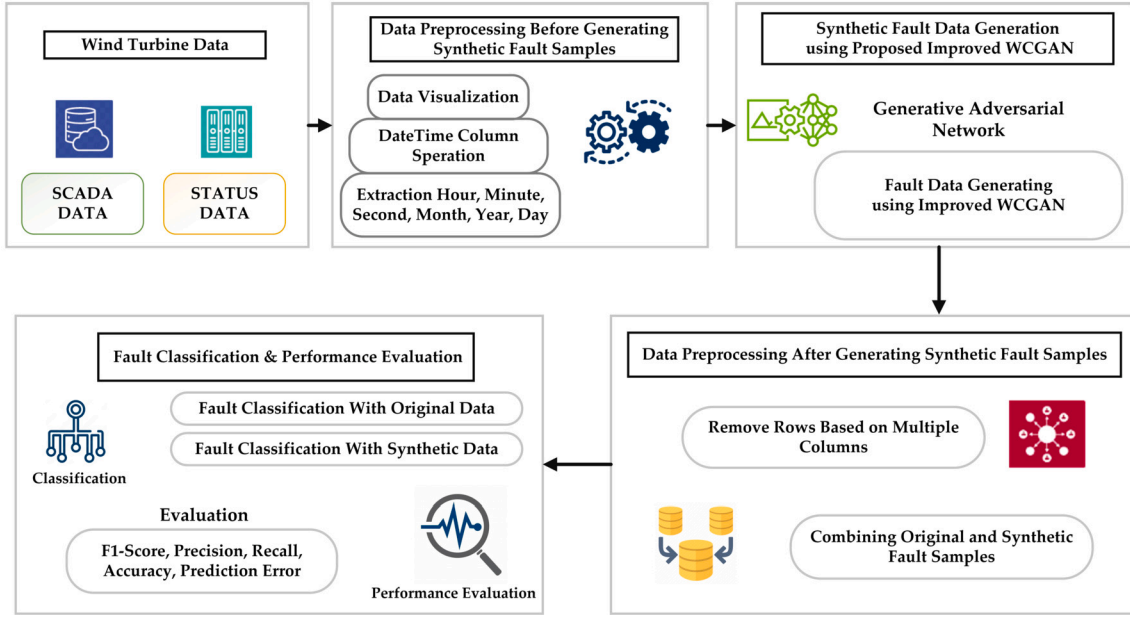


Fig. 7. Workflow of the proposed framework.

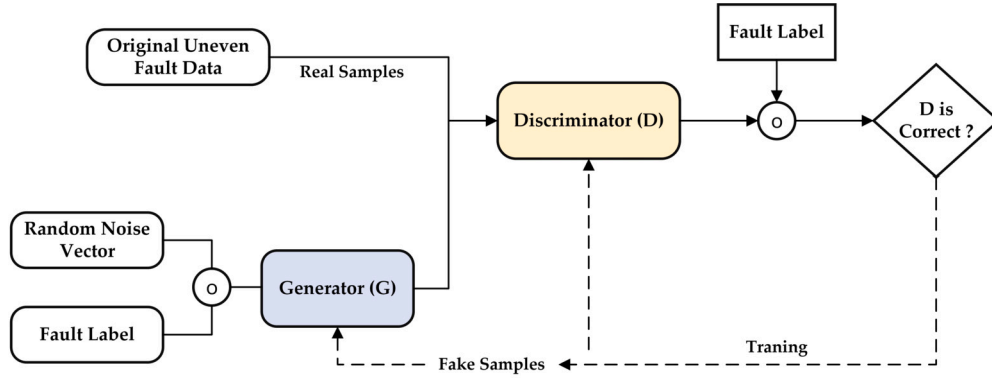


Fig. 8. Generative Adversarial Networks.

put is actual data; else, $D(x)$ should produce 0. The discriminator learns the features of the actual data during this training procedure. The training loss function is represented in the Equation (1),

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p_z} [\log(1 - D(G(z)))] \quad (1)$$

Where x denotes the actual or real data sample and z represents random noise vector. $G(z)$ represents the generated sample, while $E_{x \sim p_{data}}$ the actual data distribution and $E_{z \sim p_z}$ the fake data distribution. The parameters of the two networks are updated by alternating gradient descent. First, the generator's parameters are fixed; the discriminator is trained using the generated and actual data. Then the discriminator parameters are fixed by continuously learning from the loss function, and the generator's parameters are updated. The generator can finally generate realistic data like the original data by repeated alternating training.

The above objective function is performed by unsupervised learning with randomly generated data. Deployment is complicated when your dataset has imbalanced data issues, and the data has multiple classes that must be learned. An alternative for this problem is conditional GANs [46]. CGAN has additional label information to simplify the training process to generate monitored data. Therefore, a trained generator can create any class if the appropriate labels are provided.

The given random noise and labels are combined into an input vector for the generator G , as shown in Fig. 8. Second, discriminators are used to distinguish samples from synthetic or actual samples. If discriminator D cannot make a correct decision at this point, it is updated. The backpropagation algorithm also optimizes generator G . Eventually, both achieved the Nash Balance and were well-trained. Equation (2) can therefore be used to express the conditional probability objective function of CGAN.

$$\min_G \max_D V(G, D) = E_{x \sim p_{data}(x)} [\log D(x | y)] + E_{z \sim p_z(z)} [\log(1 - D(G(z | y)))] \quad (2)$$

The design of a detailed conditional adversarial net is shown in Fig. 9. If the generator and discriminator depend on some additional information y , which is class labels, the GAN can be expanded to a CGAN. We can achieve conditional sampling by providing y to the discriminator and generator as additional input layers.

The adversarial training framework offers much freedom in how this hidden representation is put together. In the generator, the prior input noise $p_z(z)$ and y are combined in joint hidden representation $\log(1 - D(G(z | y)))$. The discriminator gives the discriminative function representation $\log D(x | y)$ x and y as inputs.

For both standard GANs and CGANs, the KL divergence score [48] has been used to estimate the divergence between the two distributions.

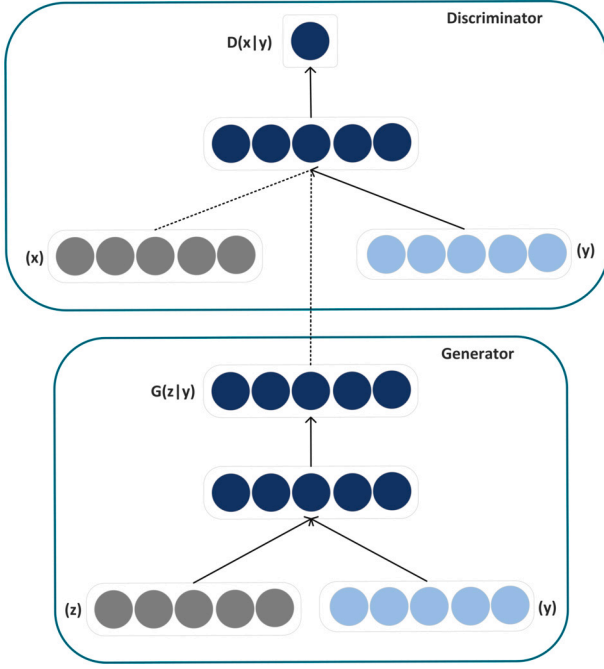


Fig. 9. Overall architecture of CGAN [47].

The KL divergence score measures the degree to which one probability distribution differs from another. Equation (3) is frequently used to describe the KL divergence.

$$KL(P\|Q) = \sum_{i=1}^N P(x_i) \log \frac{P(x_i)}{Q(x_i)} \quad (3)$$

The equation above illustrates the divergence of the two distributions, Q and P . Here, the log of the chance of each event in P over the probability of each event in Q is multiplied. This approach is the more common implementation used in practice. According to the KL divergence score's intuition, there is a significant divergence when an event's probability from P is high while its likelihood from Q is low. There is also a considerable divergence, but it is less significant than in the first situation when the likelihood from P is small and the probability from Q is excellent.

In contrast to employing KL divergence, recent studies have demonstrated improvements in data generation using Wasserstein distance. Therefore, in this study, we opted to replace the KL divergence in the CGAN with the Wasserstein distance. Because of this, the Wasserstein distance [49] explained in Equation (4) is used in this experimental investigation instead of the KL divergence, which is used for CGAN.

$$W(P_r, P_g) = \inf_{\gamma \in \Pi(P_r, P_g)} E_{(x,y) \sim \gamma} [\|x - y\|] \quad (4)$$

The common distribution r can be obtained via sampling to obtain actual samples x and produced synthetic samples y , and their distances $\|x - y\|$ can then be calculated where $\Pi(P_r, P_g)$ is a collection of joint distribution for P_r and P_g .

The Wasserstein distance has the potential to not only address the issue of unstable training but also act as a trustworthy gauge of how well a GAN is doing during training.

After weight sharing, the gradient can occasionally disappear. We use the gradient punishment (GP) method to address this issue and meet the Lipschitz continuity requirement. GP resolves the issue of training gradient disappearance or gradient explosion. It can produce more excellent-quality samples and has a faster convergence rate. As a result, the objective function of taking GP into account can be stated as

Table 4
Hyperparameter settings for generator network.

Parameters	Generator
Learning rate	0.0001
Hidden Layer Optimizer	ReLu
Output Optimizer	Adam
Loss function	Wasserstein distance + GP
Dropout	0.2
Random Noise	50

Table 5
Hyperparameter settings for discriminator network.

Parameters	Discriminator
Learning rate	0.00001
Hidden Layer Optimizer	ReLu
Output Optimizer	Adam
Loss function	Binary Cross Entropy
Dropout	0.2

$$L = E_{x \sim P_g} [D(x)] - E_{x \sim P_r} [D(x)] + \lambda \cdot E_{x \sim P_g} \left[\left(\|\nabla_{\hat{x}} D(G(z))\|_p - 1 \right)^2 \right] \quad (5)$$

The underlined portion of Equation (5) represents the gradient penalty term of GP. Combining Equations (2) and (5) produces the optimal function of improved adversarial learning shown in Equation (6).

$$\min_G \max_D V(G, D) = E_{x \sim P_{\text{data}}(x)} [\log D(x | y)] + E_{z \sim P_z(z)} [\log(1 - D(G(z | y)))] + \lambda \cdot GP \quad (6)$$

Data from the original fault samples were utilized to train the model. The improved WCGAN processes the fault samples using the Wasserstein distance and GP technique. Tables 4 and 5 lists the generator and discriminator model's hyperparameter settings. As shown in Tables 4 and 5, the learning rates for constructors and discriminators are 0.0001 and 0.00001, respectively. The generator's input is the random noise distribution $z(50)$. The Adam optimizer was used to manage this optimization learning process. The gradient penalty (GP) parameter λ is set to 1, dropout to 0, and number of epochs to 700. The generator training rule was trained once, and the discriminator training rule was taught five times per epoch.

4.2. Data post preprocessing after generating synthetic fault samples

After generating synthetic data, data post-preprocessing is an essential step in this experimental setup. After sampling data that only generated labels 62,80,228,60 and 9, we have to trim the synthetic data as per the original data "Date" distribution. We combine year, month, and day columns into a single date column. We combine the hour and minute columns into a single Time column and finally merge the Time and Date columns into one single Time column to match the SCADA data Time column format. We dropped the extra columns which are not required for the final data. The final dataset is named final_data. Filter rows by dates in the data frame to match the date format with the original data. We have removed rows based on the month format and removed generated extra dates that do not belong to April, May, June, September, and November.

Fig. 10 shows the detail of the fault data generated from five fault classes, where fault 62 has the highest samples and fault class 228 has the lowest. We have overcome the highly imbalanced data problem by generating synthetic fault samples.

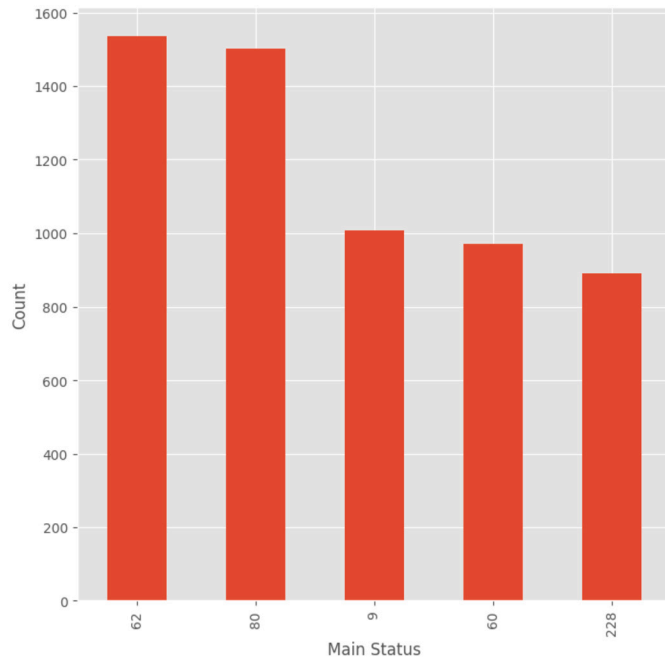


Fig. 10. Visualization of generated fault data samples for five fault classes.

4.3. Analysis of applied ensemble model for the fault classification

An ensemble learning method reduces the potential deviation of a single classifier in handling imbalanced data by training many weak classifiers. Bagging, boosting, and stacking are the three essential components of ensemble learning. Developing an ensemble or tree-based classification framework is the basis of most methods for dealing with imbalanced data classification. This study employed a stacking technique to mitigate the challenges of poor generalization and an elevated misclassification rate associated with a single classifier. This approach proves particularly advantageous in dealing with imbalanced datasets.

This experiment utilized an ensemble stacking classifier for an extended classification task, incorporating three distinct models for prediction. The subsequent aggregation of predictions from these three models employed a random forest as a meta-model. This advanced approach aims to leverage the diverse strengths of individual models, enhancing the overall robustness and accuracy of predictions. The ensemble stacking classifier enriches the decision-making process by incorporating collective insights from multiple models. Moreover, the integration with a random forest meta-model serves to further refine predictive performance, providing a comprehensive and synergistic solution for tackling imbalanced data classification tasks. Fig. 11 illustrates the stacking approach. The Extra trees Classifier, logistic regression, and XGBoost are selected as base learning models. Random Forest is utilized as the meta-learner model for aggregating.

Stacking ensembles combine the predictive strengths of multiple base models to perform better than individual models in classifying imbalanced multiclass data. Accumulation adds predictions and can reduce errors due to overfitting and model bias. In contrast to logistic regression, XGBoost performs well with imbalanced datasets by increasing weights on minority classes while logistic regression excels in separating linearly separable classes. In this way, a robust system can be developed to deal with SCADA data imbalances by combining diverse models.

In contrast to homogeneous ensembles, hybrid ensembles draw on the unique strengths of a variety of models, giving it an edge over homogeneous ensembles. This study shows that a classifier based on extra trees, a logistic regression, and an XGBoost system can be combined to handle non-linear data, regularize high-dimensional data, and mitigate imbalances. This fault classification can be enhanced with Random Forest, which aggregates these predictions. In SCADA, homogeneous

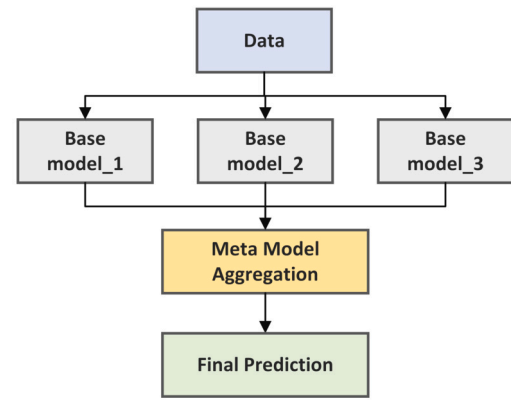


Fig. 11. Developing an ensemble classifier involves a series of steps.

ensembles fail to perform well when presented with imbalanced or complex datasets due to their lack of diversity.

The stacking ensemble in our study employs an extra trees classifier, logistic regression, and XGBoost as base learners. As a result of weighted learning, each of the models provides a unique advantage: extra trees provide robustness to class imbalances, logistic regression brings simplicity and interpretability, and XGBoost handles high variance effectively. Concerning the three models, each offers unique benefits; extra trees provide robustness to class imbalances through weighted learning, logistic regression provides simplicity and interpretability, and XGBoost provides robustness to class imbalances through logistic regression.

Using these models, predictions are generated that are then fed into a Random Forest meta-learner that captures interdependencies between predictions from the base model and outputs the outcome. As opposed to simple voting classifiers, stacking ensembles leverage patterns from multiple base models to enable nuanced decision-making.

code run and check: We have included MCC (Matthews Correlation Coefficient) in our evaluation to provide a balanced assessment of classification performance. Unlike metrics like accuracy, MCC considers true positives, false positives, true negatives, and false negatives equally, making it particularly useful for imbalanced datasets. Its inclusion demonstrates the robustness of our proposed stacking ensemble model in addressing minority-class prediction challenges effectively.

4.3.1. Extra trees classifier

An ensemble learning technique called Extremely Randomized Trees Classifier (Extra Trees Classifier) aggregates the outcomes of various decorrelated decision trees gathered in a forest to produce its classification result. Its sole conceptual distinction from a Random Forest Classifier is how the decision trees in the forest are built. The original training sample is the foundation for each Decision Tree in the Extra Trees Forest. Then, a random sample of k features from the feature set is given to each test node of each tree. According to various mathematical criteria, such as the Gini Index illustrated in Equation (7), each decision tree must choose the ideal characteristic to divide the data.

4.3.2. XGBoost

XGBoost is a machine-learning technique that sequentially uses decision trees. Weights are assigned to each independent variable before feeding it into the decision trees for predicting outcomes. If a tree inaccurately predicts a variable, its weight is incremented before being inserted into the next decision tree. This weight incrementation process helps to create a strong and accurate model ensemble, with individual classifiers working together throughout the procedure. XGBoost is a widely used and powerful gradient-boosting algorithm that boasts of speed, scalability, and robustness to overfitting. It is highly effective in feature selection and regularization and can be integrated into ensemble approaches, to achieve improved performance. Unlike the bagging technique, XGBoost uses boosting to create predictors sequentially. This

“regularized boosting” process allows for cross-validation at each iteration, making it easy to determine the optimal number of boosting iterations in a single run. In equation (7) When using p regression trees that build on the previous one, we can define the predicted target $\hat{y}$. To minimize the objective function, which includes both loss and regularization components, we use equation (8), where, first term is the loss function and the second is the regularization parameter Here, l denotes the explanation of the loss function, and Ω is the regularization function. By using XGBoost, we can achieve precise and efficient machine learning that can improve performance and prevent overfitting.

$$\hat{y}_i = \sum_{p=1}^P f_p(x_i), f_p \in \mathcal{F} \quad (7)$$

$$\text{obj}(\theta) = \sum_i^n l(y_i, \hat{y}_i) + \sum_{p=1}^P \Omega(f_p) \quad (8)$$

4.3.3. Logistic regression

Logistic regression (LR) is utilized when the target variable is categorical. LR is yet another machine learning-inspired statistical analysis technique. Decision boundaries are used to divide various sample types into groups in n -dimensional space, and cost functions relating to model parameters are used to gauge how well the model can be trained. Decision boundaries are continuously fitted to complete the classification assignment to optimize the most significant probability value. A logistic regression auxiliary classifier based on the kurtosis threshold is established in the field of fault diagnosis to achieve fault classification and gradient punishment for the discriminator. The non-normal distribution fault can be dichotomized for different gear fault samples, and the model parameters can be optimized by weight.

4.3.4. Random forest classifier

Random forests (RF) [50] is a supervised machine learning algorithm for classification, regression, and other applications. We used RF because, in contemporary machine learning, tree ensemble models like Random Forests nearly consistently outperform single-decision trees. Random forests are excellent at handling high dimensional and subsets of data. Because it uses a portion of the data in this model and can readily take hundreds of characteristics, it trains more quickly than decision trees. Many different decision trees are used in the random forest approach. The random forest method uses decision tree forecasts to choose the result. It produces estimates by averaging the outcomes from various trees. With more trees, the outcome becomes more accurate. It creates several decision trees during training and employs a majority vote to determine the outcome to improve accuracy and produce more dependable forecasts.

$$IG(N_p, a) = Gini(N_p) - \sum_{i=1}^j \frac{|N_i|}{|N_p|} Gini(N_i) \quad (9)$$

$$Gini(N_p) = 1 - \sum_{j=1}^n (p_j)^2$$

Gini index illustrated in Equation (9) to determine which is the lowest index of each branch on a node based on the class and probability, indicating which branches are more likely to occur. Here, n is the number of classes, and p_j denotes the relative frequency of the class we observe in the dataset.

5. Results

In the exploration of multi-class fault classification, accuracy has traditionally been a prevalent metric for evaluating classification problems. However, its adequacy diminishes when confronted with highly imbalanced data. The model scrutinizes the majority-class samples and often misclassifies minority-class samples as the majority. Relying solely on

accuracy in imbalanced data scenarios can yield misleading results. To address this, metrics such as precision, recall, F1-score, and Matthews Correlation Coefficient (MCC) are crucial as they provide a comprehensive evaluation of model performance, capturing nuances in classification across all classes, particularly in imbalanced scenarios. To obtain a more accurate evaluation, we opted for a confusion matrix, a graphical representation that summarizes the prediction outcomes of a classification task. The matrix displays accurate classifications on the diagonal and misclassifications for each class in the corresponding row. Subsequently, accuracy, precision, recall, and F1-score measures are computed. Machine learning systems typically demand abundant data for comprehensive classification and detection strategies. Addressing the challenge of class imbalance, we employed a GAN for data augmentation, generating synthetic fault data. The generated data is then split into training and testing sets, with an 80-20 ratio. The results section encompasses a comparative analysis of various classification methods, evaluating outcomes based on F1-score, precision, recall, and accuracy. In this study, we initially generated wind turbine fault data using a GAN-based method, trained an ensemble fault detection model with synthetic fault data, and executed wind turbine fault detection using the trained ensemble model.

Accuracy: It is the percentage of accurately predicted outputs. To assess the model's dependability, it counts the number of positive and negative data that were correctly categorised. The mathematical expression is as follows (10).

$$\text{Accuracy} = (t_p + t_n) / (t_p + f_p + t_n + f_n) \quad (10)$$

Recall: A measure of a classifier's completeness. This means it gives a proportion of actual positives that are correctly classified. The mathematical expression is as follows (11).

$$\text{Recall} = t_p / (t_p + f_n) \quad (11)$$

Precision: A measure of a classifier's exactness and the quality of a positive prediction made by the model. The mathematical expression is as follows (12).

$$\text{Precision} = t_p / (t_p + f_p) \quad (12)$$

F1-Score: The F1 score is the weighted average of recall and precision. Therefore, both false positives and false negatives are considered while calculating this score. The following is the mathematical expression (13).

$$F1 - \text{score} = 2t_p / (2t_p + f_p + f_n) \quad (13)$$

where t_p denotes True Positive, t_n denotes True Negative, f_n denotes False Negative, and f_p denotes the False Positive.

G-mean: In the Geometric Mean (G-mean), both majority and minority classes are weighed equally for classification performance.

$$G - \text{mean} = \sqrt{\text{Precision} * \text{Recall}} \quad (14)$$

5.1. Implementation environment details

For the classification task in our experiment, we employed the system specification displayed in Table 6. The experiments were carried out and implemented using Python as the primary language. Using the operating system Ubuntu 20.04.2 LTS and an Intel(R) Core(TM) i7-8700 clocked at 3.20 GHz, we tested our method with GeForce RTX 2080 SUPER as a GPU.

5.2. Comparative experiments of wind turbine Fault classification

This section elaborates on the four groups of comparative experiments that are detailed below to test the fault classification of the proposed method.

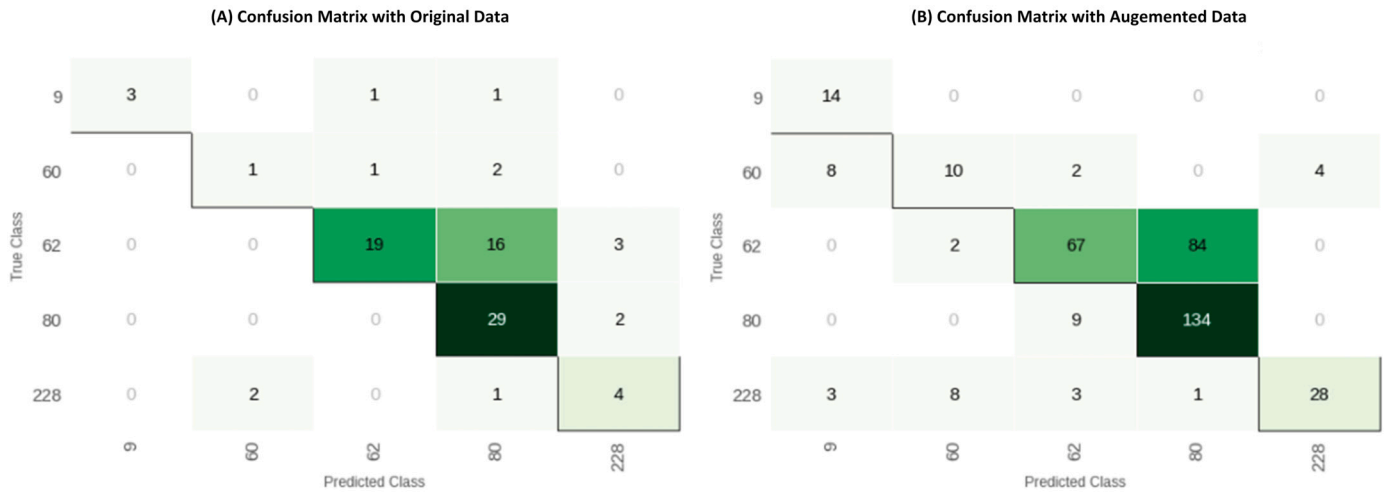


Fig. 12. (A) Visualization of confusion matrix with original data, (B) showing the confusion matrix with merged original and generated synthetic data.

Table 6

Implementation environment for the experiment.

System Specification	Description
Operating System	Ubuntu 20.04
CPU	Intel(R) Core (TM) i7-8700
RAM	32 GB
GPU	GeForce RTX 2080 SUPER
Programming-Language	Python
IDE	Jupyter Notebook

5.2.1. Comparative experiments using the original fault Data and the generated fault data

In Fig. 12, the confusion matrix shows the class labels, where label 0 denotes class 9, label 1 denotes class 60, label 2 represents class 62, label 3 marks class 80, and label 4 indicates class 228. It is noticeable that after applying our approach to the imbalanced wind turbine fault data, the performance improved, with labels being more correctly classified compared to the original data. Additionally, when the F1-score value is at 73.01%, the values for recall and precision are higher than 68.7% and 68.6%, respectively, indicating that the majority of the samples have been correctly identified. The study's findings demonstrate that GAN-based data augmentation for balancing the data and the ensemble approach achieved low misclassification outcomes and high prediction rates.

When using the original imbalanced fault data for classifying faults, the accuracy obtained could have been better at 63%, recall at 57%, precision at 61%, and F1-score at 61%. The recall and precision are low, indicating they pull down the F1-score. After that, we employed GAN-based augmentation to overcome our experiment's imbalanced fault data problem. Utilizing the combination of balancing the dataset improved classification results. The highest performance achieved by the model in accuracy was 68%, recall 68%, precision 68%, and F1-score 73%.

In the analysis presented in Fig. 12, the confusion matrix provides a visual representation of the performance of the proposed model. The vertical axis (y-axis) represents the actual faults, while the horizontal axis (x-axis) displays the faults predicted by the model. The diagonal of the matrix indicates instances where the predictions align with the actual faults. In Fig. 12 (A), which reflects the classification results using the original data, each cell in the matrix corresponds to a specific fault class. For instance, Class 9 has 5 samples, and the model predicts 3 correctly, but 2 are misclassified as Class 62 and 80. The overall accuracy on imbalanced data is 63%, but the matrix reveals challenges in accurately predicting various fault classes, especially due to class imbalance. Fig. 12 (B) presents the classification outcome using augmented

data. Here, Fault 9 is predicted with 100% accuracy, demonstrating the positive impact of the augmented dataset. However, other faults exhibit a mix of accurate and inaccurate predictions. For example, Fault 60 has 24 samples, with 10 predicted accurately and 14 misclassified. Comparing Figs. 12 (A) and (B), it becomes evident that augmenting the original data with GAN-generated synthetic fault data significantly enhances the model's ability to correctly classify faults. The augmentation results in an increased number of correctly classified samples, contributing to a 5% improvement in overall accuracy for the classification technique. This suggests that the integration of synthetic data plays a crucial role in addressing challenges associated with imbalanced fault data and improving the model's predictive performance.

Table 7 provides a comprehensive overview of the accuracy comparison results derived from our study, shedding light on the impact of data augmentation on addressing classification challenges within imbalanced datasets. The table highlights that when our proposed model is utilized for classifying both the original imbalanced data and the augmented balanced data, superior precision, recall, and F1 scores are achieved, particularly with the augmented dataset exhibiting expanded proportions. The overall accuracy across all models is summarized in Table 7, with the stacking classifier emerging as the top performer, boasting an accuracy of 68% alongside commendable precision, recall, and F1-score values. This reinforces the reliability of our classification accuracy and underscores the efficacy of a GAN-based synthetic data generation model in mitigating data imbalance challenges.

As one of the performance metrics, we used the MCC to evaluate the empirical validity of our approach. A balanced measure of classification performance is provided by MCC when used with imbalanced datasets. A perfect classification is represented by a score of +1, and a perfect misclassification is represented by a score of -1.

We evaluated several classifiers before and after applying the augmentation technique. The results showed that MCC scores improved consistently across all classifiers, highlighting the effectiveness of the data-augmented training models. Detailed MCC results are presented in Table 8. Although the proposed approach has helped to some extent in correcting the imbalance issue, there is still room for improvement to further improve the classification results. As part of our future research, we will explore and incorporate advanced methods and alternative sampling techniques, thereby improving performance.

Fig. 13 further delves into the nuances of multiclass classification prediction errors, showcasing the differences after amalgamating synthetic and original data. In the context of our multiclass fault classification approach, where each class has varying sample sizes, GAN-based data augmentation contributes to a reduction in prediction errors compared to the original data. The stacked bar chart in Fig. 13 illustrates the Class Prediction Error of our suggested classifier, offering insights into

Table 7
Performance comparison with different classifiers model.

Classifier	Evaluation Metrics With Original Data				Evaluation Metrics With Augmented Data			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
Decision Tree Classifier	0.48	0.48	0.47	0.48	0.62	0.63	0.62	0.62
SVM - Linear Kernel	0.50	0.46	0.50	0.42	0.59	0.64	0.59	0.55
Ridge Classifier	0.52	0.51	0.52	0.51	0.62	0.63	0.62	0.61
Gradient Boosting Classifier	0.52	0.52	0.52	0.51	0.64	0.64	0.63	0.64
Light Gradient Boosting Machine	0.54	0.55	0.54	0.54	0.64	0.64	0.64	0.64
Extreme Gradient Boosting	0.56	0.56	0.56	0.55	0.63	0.64	0.63	0.63
Voting Classifier	0.58	0.58	0.58	0.54	0.65	0.66	0.65	0.65
Proposed Stacking Classifier	0.63	0.61	0.57	0.61	0.68	0.68	0.68	0.73

Table 8
MCC Scores Before and After Augmentation.

Classifier	MCC Before Augmentation	MCC After Augmentation
Decision Tree Classifier	+0.41	+0.57
SVM - Linear Kernel	+0.44	+0.52
Ridge Classifier	+0.48	+0.58
Gradient Boosting Classifier	+0.47	+0.57
Light Gradient Boosting Machine	+0.50	+0.59
Extreme Gradient Boosting	+0.53	+0.61
Voting Classifier	+0.55	+0.63
Proposed Stacking Classifier	+0.59	+0.66

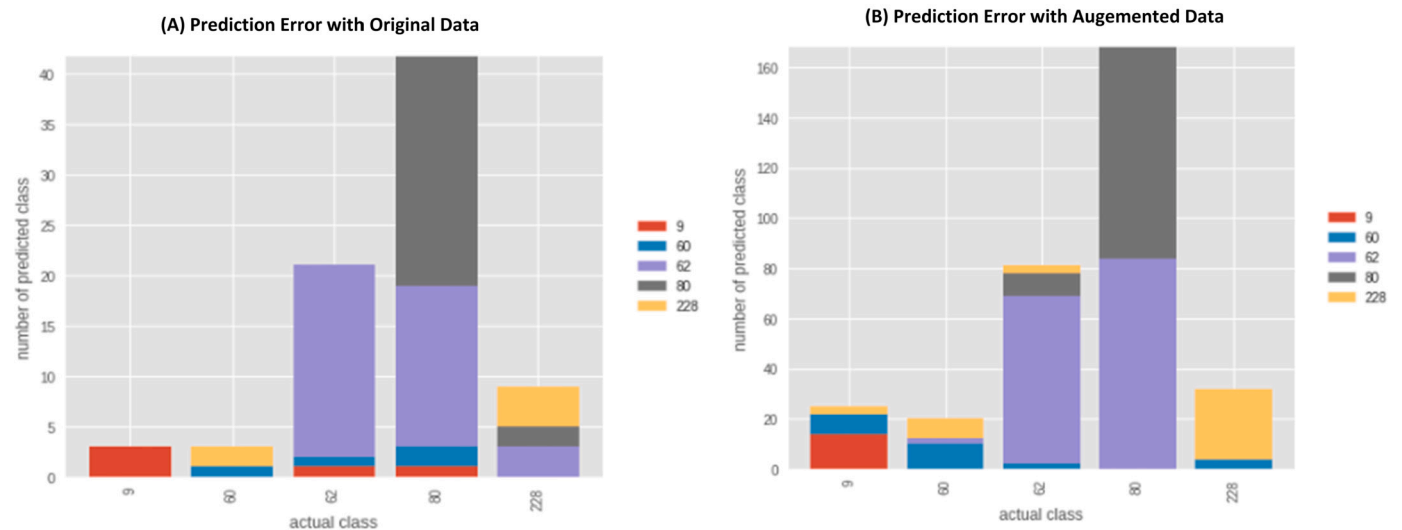


Fig. 13. Prediction error comparison results of fault detection using original fault data and generated synthetic fault data. (A) original data, (B) merged original and synthetic augmented data.

the classes posing challenges for the classifier and highlighting individual inaccuracies. Remarkably, class 9 exhibits zero incorrect classifications, while other classes (60, 62, 80, and 228) display some inaccurate predictions. Notably, all classes experience some degree of incorrect predictions, though with minimal errors, as evidenced by the bar chart on the right side.

To further assess the model's performance, ROC curves are employed, providing a visual representation of its diagnostic efficacy across various thresholds and presenting trade-offs between true positive and false positive error rates. The ROC plot serves as a valuable tool for evaluating the classifier's performance in multiclass classification scenarios, offering a nuanced analysis of sensitivity and specificity values.

The ROC metrics, often referred to as sensitivity and specificity, play a pivotal role in assessing the performance of a classification model. Sensitivity denotes the model's ability to correctly identify entries belonging to the positive class, while specificity measures its accuracy in

recognizing entries from the negative class. In the realm of multi-class fault classification, the ROC curve, contrasting each fault with others, is meticulously presented in Fig. 14. The Y-axis represents the true positive rate, while the X-axis illustrates the false positive rate. The ideal point, characterized by a false positive rate of zero and a true positive rate of one, resides in the upper-left corner of the graph. Thus, an effective classifier manifests an ROC curve that closely aligns with this optimal point, indicating precise diagnostic test support.

Fig. 14 (A) vividly portrays the ROC curves of the classifier model, showcasing their orientation away from the upper-left corner. Noteworthy is the exemplary performance of fault 9, achieving a perfect 100%, while other faults exhibit varying degrees of predictive values. The curves in Fig. 14 (B) exhibit a notable inclination towards the upper-left corner, with faults 9 and 228 demonstrating proximity, and fault 62 displaying the lowest predictive ability across all values compared to its counterparts. Despite these variations, the classifier maintains an im-

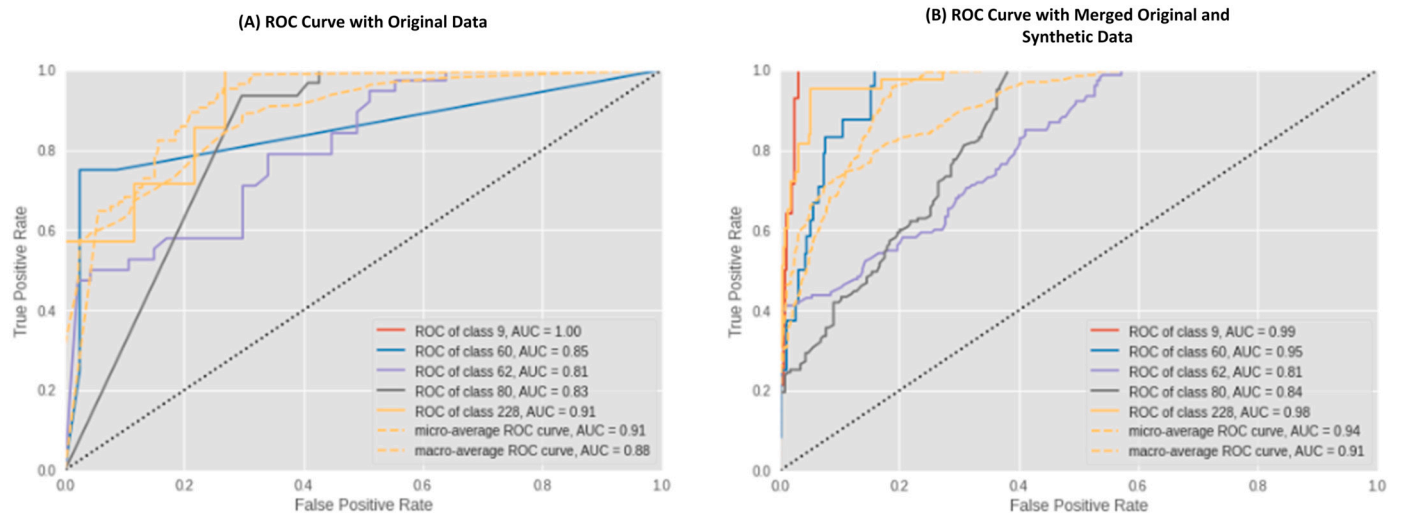


Fig. 14. ROC score performance for multi-class classification with (A) original data, (B) merged original and synthetic augmented data.

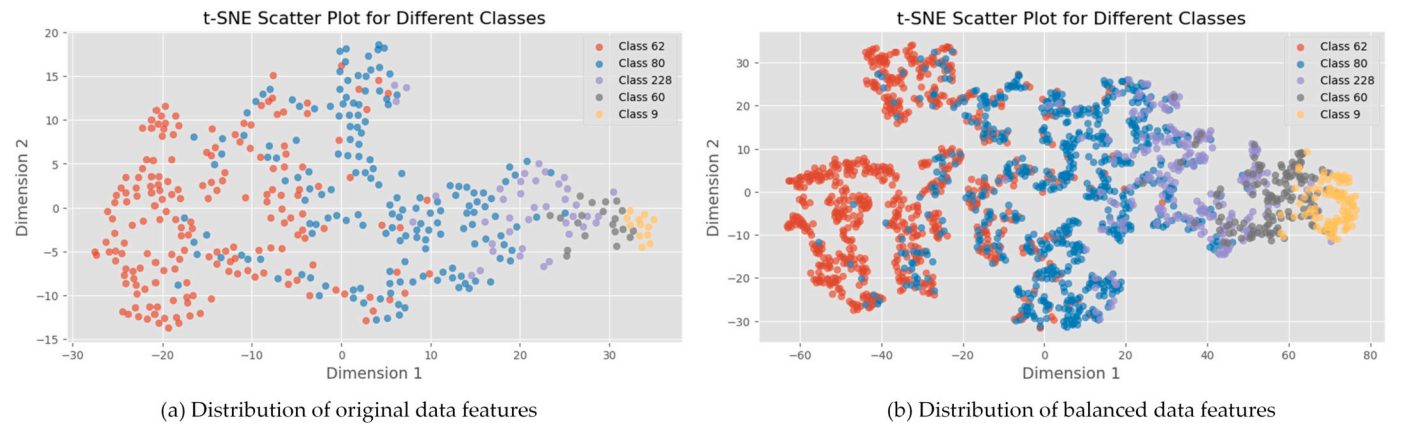


Fig. 15. t-SNE visualization of fault data distribution: (a) original data features, (b) balanced data features using proposed GAN technique.

pressive level of accuracy across an extensive spectrum of sensitivity and specificity values, underscoring its robust diagnostic capabilities. This nuanced analysis of the ROC curves not only offers a comprehensive understanding of the model's discriminatory prowess but also emphasizes its reliability in discerning faults with varying predictive values, contributing to the overall efficacy of the classification system.

5.2.2. Comparative analysis of original and generated fault data

Clustering performance can be used to assess the quality of generated fault features. However, it hides outliers because it doesn't directly show how separate clusters are. To address this, we employed t-SNE (t-Distributed Stochastic Neighbor Embedding) [51] to visualize the high-dimensional data in a two-dimensional space, as seen in Fig. 15. The t-SNE plot helps us see how well-separated the different fault classes are. Fig. 15(a) shows the fault features for the unbalanced dataset. Notice the uneven distribution of data points across different classes. Fig. 15(b) demonstrates the balanced dataset after using our proposed GAN model. Here, we can see a clearer separation between the classes, the dots in the plot represent various fault classes: red for class 62, blue for class 80, purple for class 228, grey for class 60, and yellow for class 9. Due to the more balanced distribution achieved through our technique. Balancing the data leads to better separation of fault features and improves classification accuracy.

5.2.3. Comparative experiment with the generated fault data

Table 9 offers a comprehensive comparison between our proposed fault classification model and established methods applied to the same

dataset. It evaluates various metrics, including accuracy, the number of fault classes considered, precision, recall, and the necessity for data balancing. The comparison encompasses a range of deep learning techniques such as LSTM, CNN-LSTM, CNN-GRU, and neural networks, as well as shallow machine learning methods like SVM, DT, and RF. Notably, the analysis reveals that relying solely on overall accuracy is inadequate for validating balanced classification, particularly given the limited and imbalanced nature of fault data in wind turbine datasets. As a result, achieving high classification accuracy with different well-known machine learning models proves challenging due to the small and imbalanced dataset size.

5.2.4. Comparative study using a small number of actual fault data samples

This comparative experiment aims to assess the accuracy of fault detection using a limited number of samples. The fault detection model is specifically engineered to identify faults occurring when the wind speed is 9 m/s. However, the available data for training the wind turbine fault classification model is insufficient due to the scarcity of real fault data samples in such conditions. Consequently, the models trained with this small dataset exhibit poor classification performance, as they struggle to detect faults effectively. To address this limitation, we integrated the generation of synthetic fault data and trained the model using a combination of original and synthetic data, thereby augmenting the dataset for training. Despite the additional synthetic data, the generated fault data alone are inadequate for fault detection. However, we observed a significant improvement upon comparing the performance of the mod-

Table 9

This table compares the performance of the proposed fault classification model with established methods applied to the same dataset.

Model	Number of fault classes	Accuracy	Precision	Recall	F1-score	The requirement for data balancing
LSTM	5	62.59	53.61	35.65	35.34	Yes
CNN-GRU	5	64.08	58.55	40.54	39.05	Yes
CNN-LSTM	5	62.32	60.84	40.85	41.22	Yes
DT	5	61.54	62.24	38.29	38.75	Yes
SVM	5	64.65	62.29	61.85	62.29	Yes
RF	5	65.75	64.94	64.95	65.31	Yes
Proposed Classifier	5	68.67	68.64	68.75	73.01	Yes

Table 10

Performance Comparison of Data Augmentation Techniques Using Various Models.

Model	Accuracy(%)	G-mean	F1-score
CI-VAE	61.02	61.44	61.52
VAE	61.80	60.76	61.60
CVAEs	62.69	62.44	62.52
LSGAN	64.28	63.88	64.13
WGAN	66.52	66.15	67.41
CGAN	68.67	74.00	73.01

els trained with the original limited fault data to those trained with the augmented dataset. All three fault detection methods achieved accuracies exceeding 68% after being trained with the merged dataset. This outcome suggests that even in scenarios where the number of real fault data samples is minimal, the inclusion of generated fault data can substantially enhance the accuracy of wind turbine fault detection.

The classification accuracy varies when current data augmentation techniques, like Variational Autoencoders (VAE) [52,53], Conditional VAEs (CVAEs) [54], Wasserstein GAN (WGAN) [55], CI-VAE, and least squares GAN (LSGAN) [56], are used on the original dataset. In comparison, WGAN generated better synthetic data as compared with LSGAN. When we applied CGAN for synthetic data generation we achieved considerable results. By comparing the classification boundary with different source-generated data, we conclude there is a difference in distance between them. Five repeat experiments are carried out for this example to assess the generalizability, stability, and reliability of the effect of various data augmentation techniques. Table 10 lists the accuracy, G-mean, and F1-score formulas. CGAN can stabilize the data augmentation of the GAN training process. Still, the average effect of variations of VAE models, WGAN and LSGAN data augmentations is lower than that of CGAN and exhibits significant variance among tests. According to the average of the experimental results, the proposed GAN model's classification accuracy is close to 69%, and the G-mean and F1-score also significantly improve. This indicates that the collaboration and competition of generated data can further enhance the effect of GAN-based data augmentation.

6. Discussion

A significant challenge for fault classification in this dataset is the extreme imbalance present. This influences the accuracy of the proposed classification method. Despite the fact that our method shows improved accuracy over baseline approaches, the data set is inherently complex. Although it is well known that GANs are effective in resolving data imbalances, we encountered significant problems in our experiments, particularly due to extreme class imbalances in the dataset. Generating synthetic data helped improve the representation of minority classes, but training a classification model with synthetic data alone did not significantly improve performance. The model performed best when synthetic fault data was combined with the original dataset, combining the strengths of both data sets. Despite extensive experiments with different hyperparameter settings, such as learning rate, batch size, and

different epochs, the observed improvements were not significant. This result highlights that synthetic data alone is not always sufficient to resolve imbalance problems, especially when the data set is highly skewed.

To evaluate the quality of the synthetic samples generated for addressing the data imbalance issue, we employed two key metrics: the discriminant score and the coverage metric. In this study, the discriminant score, which measures how well the classifier distinguishes between real and synthetic data, was 0.65. As a result, the classifier has moderate accuracy when separating the two datasets, suggesting that real and synthetic data are somewhat similar. A coverage metric of 0.852 was also obtained, indicating that the generated data covers the modes of the real data distribution. As a result of these results, it may be concluded that synthetic samples meet acceptable quality criteria, balancing similarity and diversity. To mitigate the risks of overfitting and sample redundancy, the augmentation ratio should be fine-tuned.

This result highlights the importance of combining synthetic and real data and carefully tuning the models to achieve strong classification performance. Future research could explore advanced GAN architectures or alternative synthetic data generation techniques to further address these issues related to imbalanced data sets.

7. Conclusion

In conclusion, the efficient operation of wind turbines, vital for renewable energy generation, hinges on timely defect identification and repair. Recent studies have explored diverse classification techniques for detecting faults in wind turbine data, with a critical emphasis on addressing the challenge of imbalanced data. The imbalance in wind turbine data poses a significant hurdle to fault diagnosis performance, necessitating innovative solutions. This study introduces a GAN-based data enhancement method, leveraging the Wasserstein distance for generating suitable fault samples within the gradient punishment-based CGAN model. The experimental analysis validates the effectiveness of this approach, yielding anticipated results and contributing to increased classification accuracy. A key highlight is the utilization of a stacking ensemble approach, coupled with a generic classification method, to produce accurate findings for all power measurements. The proposed approach adeptly tackles the imbalanced samples issue by generating high-quality synthetic fault data, showcasing its potential for wind turbine fault classification. However, it's crucial to acknowledge the limitation of this research, which warrants testing the proposed approach across diverse imbalanced datasets to establish its credibility. In summary, this research presents a comprehensive and innovative approach to addressing imbalanced data challenges in wind turbine fault classification, opening avenues for advancements in renewable energy system reliability and performance. Future work is envisioned to involve the evaluation of the proposed WCGAN on various public and private datasets, comparing its performance with an ensemble classifier. Additionally, the inclusion of the no-fault class in the data and the repetition of the experiment will offer insights into the effectiveness of class imbalance approaches like oversampling and undersampling when combined with WCGAN for enhanced data augmentation. As we move forward, the network architecture established in this study holds promise for wind

turbine fault diagnosis, and ongoing efforts will focus on further refinements and improvements to the WCGAN model.

Acronyms Table

Acronym	Definition
WT	Wind Turbines
AI	Artificial Intelligence
ML	Machine Learning
GAN	Generative Adversarial Networks
CGAN	Conditional Generative Adversarial Networks
SCADA	Supervisory Control and Data Acquisition
MCC	Matthews Correlation Coefficient
WCGAN	Wasserstein Conditional Generative Adversarial Network
KL Divergence	Kullback–Leibler divergence
CMS	Condition Monitoring Systems
DNN	Deep Neural Network
SMOTE	Synthetic Minority Oversampling Technique
DA	Deep Autoencoder
IoT	Internet of Things
MAML	Model Agnostic Meta-Learning
PSO	Particle Swarm Optimization
QSCGAN	Quick Self-attention Convolutional Generative Adversarial Network
CVAE	Conditional Variational Autoencoder
ACGAN	Auxiliary Classifier GAN
CNN	Convolutional Neural Network
ADASYN	Adaptive Synthetic
LSGAN	Least Squares GAN
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
GRU	Gated Recurrent Units
SVM	Support Vector Machines
DT	Decision Tree
XGBoost	Extreme Gradient Boosting
ETC	Extra Trees Classifier
LR	Logistic Regression
RF	Random Forest
NN	Neural Networks
t-SNE	t-distributed Stochastic Neighbor Embedding
AUC	Area Under the Curve
ROC Curve	Receiver Operating Characteristic Curve
CPU	Central Processing Unit
RAM	Random-Access Memory
GPU	Graphics Processing Unit
IDE	Integrated Development Environment
ReLU	Rectified Linear Unit
GP	Gradient Penalty
KW	kilowatts
M/S	Meters Per Second

CRediT authorship contribution statement

Subhajit Chatterjee: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yung-Cheol Byun:** Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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